

Note 3.1, 3.2 The Derivatives

Definition

- If the function $y = f(x)$ is defined on a neighborhood of x_0 , and the limit

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists, then we call $f(x)$ is differentiable at x_0 , and the limit value is the derivative of $f(x)$ at x_0 , denoted by:

$$y'|_{x=x_0} = f'(x_0) = \left. \frac{dy}{dx} \right|_{x=x_0} = \left. \frac{df(x)}{dx} \right|_{x=x_0} \neq \frac{df(x_0)}{dx} = 0$$

- Remarks:**

- If the above limit doesn't exist, then we call $f(x)$ is nondifferentiable at x_0 .
- If $f(x)$ is differentiable at each x in the (a, b) , then $f(x)$ is differentiable over (a, b)
- The derivative can be interpreted as the slope of the tangent line, instantaneous rate of change and velocity.

- Theorem 1:**

If the function $f(x)$ is differentiable at $x = c$, then it is also continuous at $x = c$. (converse is false)

- Corollary:

If $f(x)$ is discontinuous at $x = c$, then the derivative of f at $x = c$ doesn't exist.

Differentiable on an Interval; One-Sided Derivatives

- Definition:**

A function $y = f(x)$ is differentiable on a closed interval $[a, b]$ if it is differentiable on (a, b) and if $\lim_{h \rightarrow 0^+} \frac{f(a+h)-f(a)}{h}$, $\lim_{h \rightarrow 0^-} \frac{f(b+h)-f(b)}{h}$ exist at endpoints.

- Strategies to judge a function failed to be differentiable:**

- The graph of f has a "corner" or "kink" in it (a sharp point)
- discontinuity
- a vertical tangent line, that is f is continuous at a and

$$\lim_{x \rightarrow a} |f'(x)| = +\infty$$